ForensicsDetective

Assignment 2 — EAS 510: Basics of AI

1. Executive Summary

PDF files have unique binary patterns depending on which application created them. By converting PDF binary data into grayscale images, classifiers can tell apart documents made by Microsoft Word, Google Docs, or Python's ReportLab. This assignment applied five types of image distortions and tested how four classifiers — XGBoost, CNN, SVM, and SGD — hold up against them.

All four models did well on clean images, with accuracies between 99.1% and 99.9%. Robustness testing showed a different picture. XGBoost dropped to 48.1% on random cropping. The CNN fell to 26.3% on bit depth reduction and 35.5% on DPI downsampling. SVM was the most stable model, staying above 96% on all but one augmentation type. SGD also held up reasonably well across the board.

The main finding is that good accuracy on clean data does not mean a model will work reliably when documents are distorted. Each model fails differently, and knowing those failure modes is just as useful as the accuracy number itself.

2. Introduction

PDFs are used everywhere — legal documents, financial reports, academic papers. In forensic work, knowing which application created a PDF can matter. If a document was supposed to be original but was actually made in a different tool, that could point to tampering.

Microsoft Word, Google Docs, and Python's ReportLab all structure PDF files differently at the binary level. These differences show up even when two documents look identical on screen. When PDF binary data is converted into a grayscale image — each byte becoming a pixel — these structural differences become visual patterns a classifier can learn.

This assignment started with 894 pre-converted PNG images across three classes. Five augmentation types were applied to simulate real-world document distortions. Four classifiers were trained on original images and tested on each augmentation type to see how accuracy changes under distortion.

3. Methodology

3.1 Dataset

Class	Source Application	Label	Count
Word	Microsoft Word	0	398
Google Docs	Google Docs	1	396
Python	Python / ReportLab	2	100
Total			**894**

Table 1: Dataset class distribution.

Word and Google Docs each have roughly four times more images than Python. Stratified splitting was used to preserve class proportions, giving 715 training and 179 test images.

3.2 Augmentation Techniques

Augmentation	Parameters	What It Simulates
Gaussian Noise	sigma 5-20	Sensor noise, scanning errors
JPEG Compression	quality 20-80	Lossy compression when sharing documents
DPI Downsampling	300 to 150 or 72 DPI	Lower-resolution scanning
Random Cropping	1-3% per border	Scanning misalignment
Bit Depth Reduction	8-bit to 4-bit	Format conversion reducing intensity levels

Table 2: Augmentation types. Final dataset: 6,258 images (6x original).

3.3 Classifiers

XGBoost: Images resized to 64x64, flattened to 4,096 features, normalized to [0,1]. 100 estimators, max depth 6, learning rate 0.1. **CNN (PyTorch):** 3-block conv architecture on 128x128 images, trained with Adam on MPS. **SVM:** RBF kernel, C=1.0, same 64x64 preprocessing as XGBoost. **SGD:** Modified Huber loss, max_iter=1000, same preprocessing.

All four classifiers were trained on original images only. Augmented images were used for robustness testing only.

4. Results

4.1 Baseline Performance

Model	Accuracy	Precision	Recall	F1
XGBoost	99.89%	99.89%	99.89%	99.89%
SVM	99.55%	99.55%	99.55%	99.55%
SGD	99.11%	99.12%	99.11%	99.11%
CNN	99.66%	99.67%	99.66%	99.66%

Table 3: Overall baseline performance on original images.

Figure 1 shows the confusion matrices for all four models on original test images.

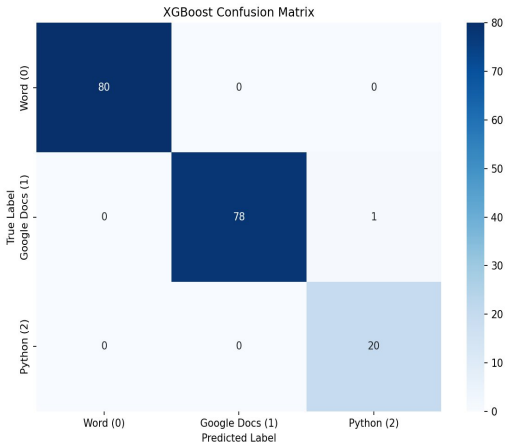

Figure 1a: XGBoost baseline (99.89%)

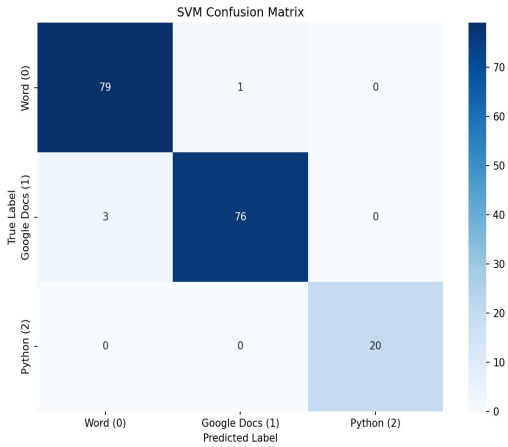

Figure 1b: SVM baseline (99.55%)

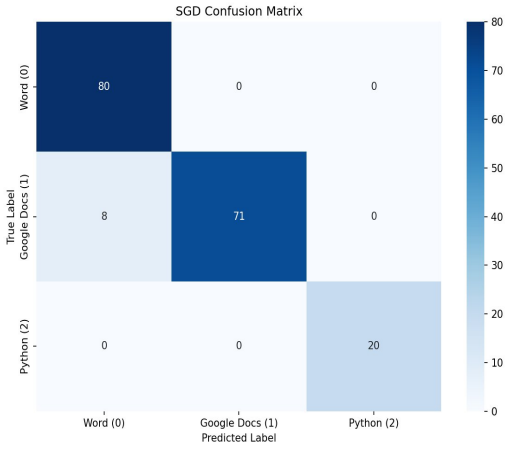

Figure 1c: SGD baseline (99.11%)

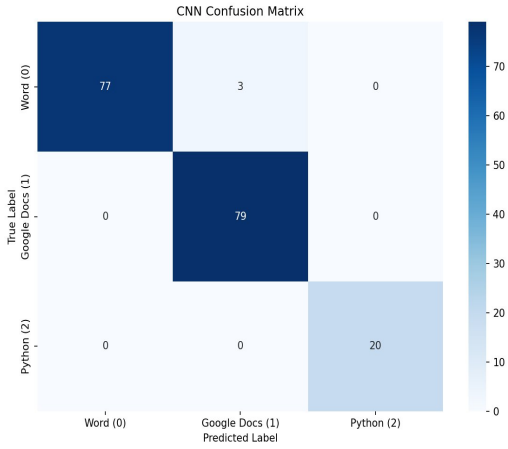

Figure 1d: CNN baseline (99.66%)

4.2 Robustness Results

Augmentation	XGBoost	SVM	SGD	CNN
Original	99.89%	99.55%	99.11%	99.66%

Gaussian	98.21%	99.55%	98.99%	99.66%
JPEG	98.21%	99.55%	99.22%	99.66%
DPI	86.13%	96.64%	93.18%	35.51%
Crop	48.10%	73.15%	81.88%	99.33%
Bit Depth	98.99%	99.66%	99.11%	26.29%

Table 4: Overall accuracy by model and augmentation. Green=high, Yellow=moderate, Orange=poor, Red=failed.

Figure 2 plots accuracy across all augmentation types for all four models.

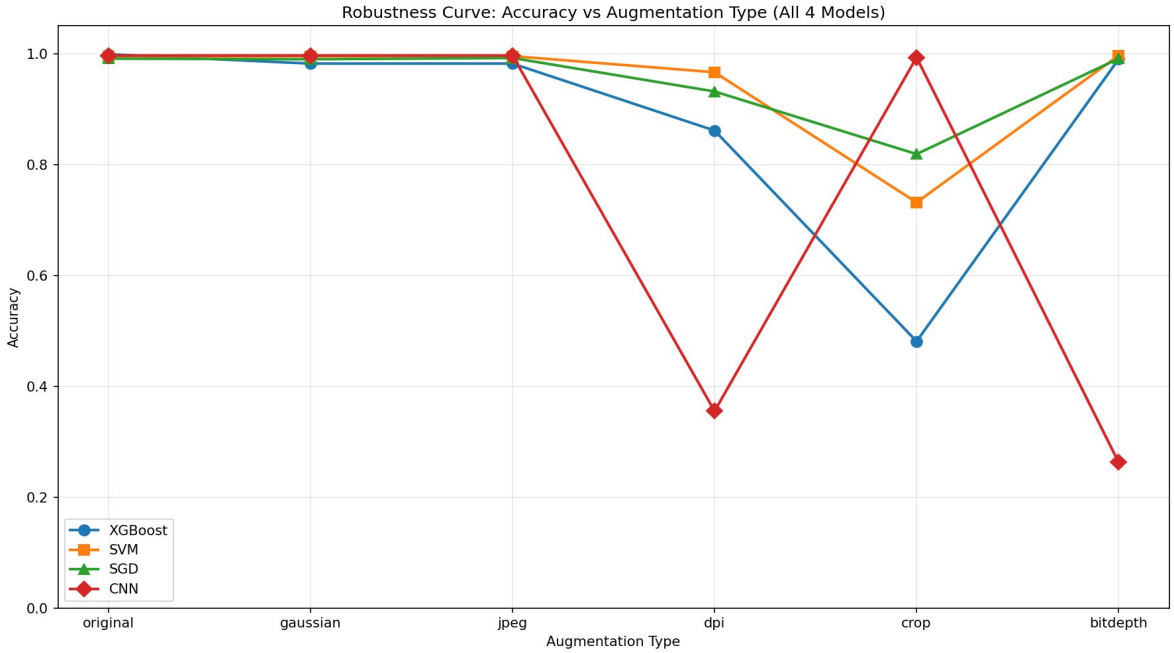

Figure 2: Robustness curve — accuracy vs augmentation type for all four models.

4.3 Per-Class Performance Across All Augmentations

XGBoost — Per-Class Results

Aug.	Class	Precision	Recall	F1
	Word	100%	100%	100%
Original	Google Docs	100%	99.75%	99.87%
	Python	99.01%	100%	99.50%
	Word	100%	100%	100%
Gaussian	Google Docs	96.34%	99.75%	98.01%
	Python	98.84%	85.00%	91.40%
	Word	99.75%	99.50%	99.62%
JPEG	Google Docs	96.81%	99.49%	98.13%
	Python	97.78%	88.00%	92.63%
	Word	98.17%	94.60%	96.35%
DPI	Google Docs	77.08%	99.37%	86.82%
	Python	0%	0%	0%
	Word	48.08%	25.13%	33.00%
Crop	Google Docs	44.69%	72.22%	55.21%
	Python	95.65%	44.00%	60.27%
	Word	100%	99.75%	99.87%
Bit Depth	Google Docs	100%	97.98%	98.98%
	Python	91.74%	100%	95.69%

Table 5: XGBoost per-class results.

SVM — Per-Class Results

Aug.	Class	Precision	Recall	F1
	Word	99.25%	99.75%	99.50%
Original	Google Docs	99.75%	99.24%	99.49%
	Python	100%	100%	100%
	Word	99.25%	99.75%	99.50%
Gaussian	Google Docs	99.75%	99.24%	99.49%
	Python	100%	100%	100%

	Word	99.25%	99.75%	99.50%
JPEG	Google Docs	99.75%	99.24%	99.49%
	Python	100%	100%	100%
	Word	95.21%	97.36%	96.27%
DPI	Google Docs	97.29%	95.08%	96.17%
	Python	100%	100%	100%
	Word	85.27%	47.99%	61.41%
Crop	Google Docs	63.68%	91.67%	75.16%
	Python	100%	100%	100%
	Word	99.25%	100%	99.62%
Bit Depth	Google Docs	100%	99.24%	99.62%
	Python	100%	100%	100%

Table 6: SVM per-class results.

SGD — Per-Class Results

Aug.	Class	Precision	Recall	F1
	Word	98.03%	100%	99.00%
Original	Google Docs	100%	97.98%	98.98%
	Python	100%	100%	100%
	Word	97.79%	100%	98.88%
Gaussian	Google Docs	100%	97.73%	98.85%
	Python	100%	100%	100%
	Word	98.27%	100%	99.13%
JPEG	Google Docs	100%	98.23%	99.11%
	Python	100%	100%	100%
	Word	87.58%	99.25%	93.05%
DPI	Google Docs	99.12%	85.35%	91.72%
	Python	98.04%	100%	99.01%
	Word	80.10%	81.91%	80.99%
Crop	Google Docs	83.20%	77.53%	80.26%
	Python	83.90%	99.00%	90.83%
	Word	98.03%	100%	99.00%
Bit Depth	Google Docs	100%	97.98%	98.98%
	Python	100%	100%	100%

Table 7: SGD per-class results.

CNN — Per-Class Results

Aug.	Class	Precision	Recall	F1
	Word	100%	99.25%	99.62%
Original	Google Docs	99.25%	100%	99.62%
	Python	100%	100%	100%
	Word	100%	99.25%	99.62%
Gaussian	Google Docs	99.25%	100%	99.62%
	Python	100%	100%	100%
	Word	100%	99.25%	99.62%
JPEG	Google Docs	99.25%	100%	99.62%
	Python	100%	100%	100%

	Word	53.68%	40.33%	46.05%
DPI	Google Docs	100%	14.39%	25.17%
	Python	18.59%	100%	31.35%
	Word	99.75%	98.74%	99.24%
Crop	Google Docs	98.75%	99.75%	99.25%
	Python	100%	100%	100%
	Word	63.64%	17.59%	27.56%
Bit Depth	Google Docs	100%	16.41%	28.20%
	Python	13.91%	100%	24.42%

Table 8: CNN per-class results.

4.4 Statistical Significance Testing

McNemar's test was used to compare classifier performance. Unlike a paired t-test on summary accuracy scores, McNemar's test works at the individual prediction level — it looks at cases where two classifiers disagree, one getting an image right while the other gets it wrong. This makes it more statistically powerful and appropriate for comparing classifiers on the same test set. Results are shown for three key augmentation conditions.

Original Images

Model Pair	chi2	p-value	Significant?	Better Model
XGBoost vs SVM	1.333	0.2482	No	-
XGBoost vs SGD	5.143	0.0233	Yes	XGBoost
XGBoost vs CNN	0.250	0.6171	No	-
SVM vs SGD	1.500	0.2207	No	-
SVM vs CNN	0.000	1.0000	No	-
SGD vs CNN	1.455	0.2278	No	-

Table 9: McNemar's test on original test images.

DPI Downsampling (hardest augmentation)

Model Pair	chi2	p-value	Significant?	Better Model
XGBoost vs SVM	120.58	< 0.001	Yes	SVM
XGBoost vs SGD	42.69	< 0.001	Yes	SGD
XGBoost vs CNN	621.46	< 0.001	Yes	XGBoost
SVM vs SGD	40.45	< 0.001	Yes	SVM
SVM vs CNN	1081.11	< 0.001	Yes	SVM
SGD vs CNN	1021.08	< 0.001	Yes	SGD

Table 10: McNemar's test under DPI downsampling.

Random Cropping

Model Pair	chi2	p-value	Significant?	Better Model
XGBoost vs SVM	171.48	< 0.001	Yes	SVM
XGBoost vs SGD	228.79	< 0.001	Yes	SGD
XGBoost vs CNN	454.02	< 0.001	Yes	CNN
SVM vs SGD	30.88	< 0.001	Yes	SGD
SVM vs CNN	232.00	< 0.001	Yes	CNN
SGD vs CNN	154.01	< 0.001	Yes	CNN

Table 11: McNemar's test under random cropping.

On clean original images, classifiers performed similarly — only XGBoost vs SGD showed a significant difference ($p=0.023$), with XGBoost performing better. Under DPI downsampling and random cropping, nearly all pairs became highly significant ($p < 0.001$), confirming that the performance differences seen in the accuracy tables are statistically real and not due to chance. SVM was significantly better than XGBoost and SGD under DPI. CNN was significantly better than all three other models under cropping. Under bit depth reduction, XGBoost, SVM, and SGD were all significantly better than CNN ($p < 0.001$), consistent with the CNN's near-complete failure on that augmentation type.

5. Robustness Analysis

5.1 Effect of Random Cropping

Cropping had the biggest split between models. XGBoost dropped from 99.89% to 48.10%, while the CNN stayed at 99.33%. SVM fell to 73.15% and SGD to 81.88%. XGBoost flattens each image into a 1D array and learns which pixel positions correspond to which class. Removing 1-3% from each border shifts all those positions. Word recall dropped to 25.13% — the model missed three out of four Word documents. The CNN handled cropping fine because its filters detect patterns across the whole image, not at fixed positions.

5.2 Effect of Bit Depth Reduction

Bit depth reduction had the opposite effect — the CNN dropped to 26.29% while XGBoost, SVM, and SGD were barely affected. Going from 256 intensity levels to 16 removes the fine texture differences the CNN relies on. The model defaulted to predicting almost everything as Python — Python recall hit 100% while precision dropped to 13.91%, and Google Docs recall fell to 16.41%.

5.3 Effect of DPI Downsampling

SVM held at 96.64% and SGD at 93.18%, while the CNN dropped to 35.51% and XGBoost to 86.13%. For the CNN, upscaling from 72 DPI creates heavy blur that looks nothing like the training data. Google Docs recall dropped to 14.39% and the model again defaulted to Python. For XGBoost, the Python class went to 0% precision and recall — class imbalance combined with DPI distortion meant XGBoost stopped predicting Python entirely.

Figure 3 shows the confusion matrices for the four most significant failure cases.

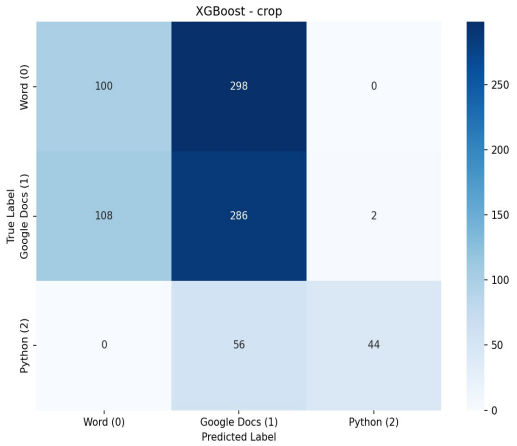

Figure 3a: XGBoost under crop (48.10%)

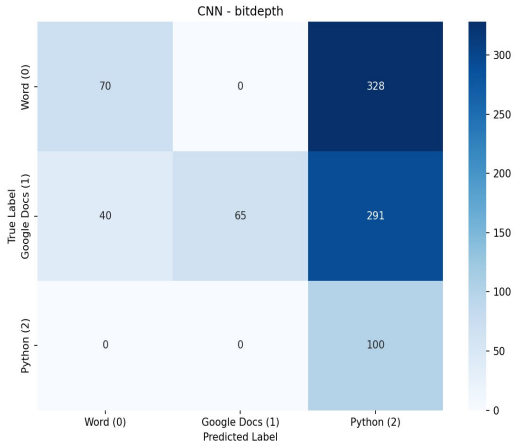

Figure 3b: CNN under bit depth (26.29%)

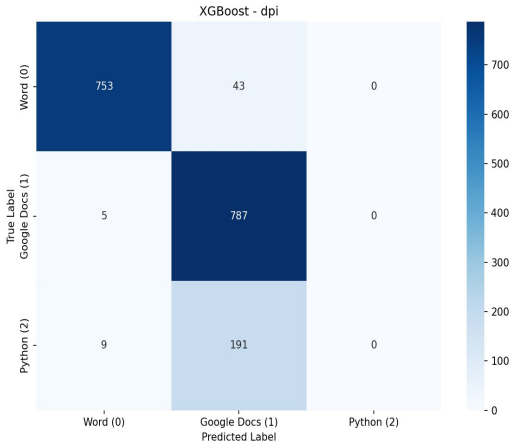

Figure 3c: XGBoost under DPI (86.13%)

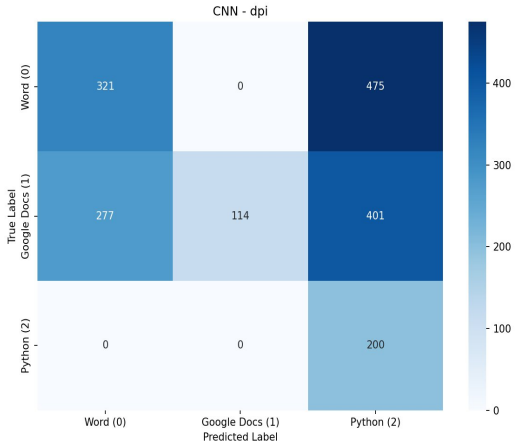

Figure 3d: CNN under DPI (35.51%)

All remaining confusion matrices are available in results/confusion_matrices/.

5.4 Gaussian Noise and JPEG Compression

All four classifiers handled gaussian noise and JPEG compression well. SVM and CNN showed no accuracy drop at all. XGBoost fell slightly to 98.21% on both, mainly because Python recall dipped under noise and compression. These distortions change pixel values but keep positions intact, which all four models tolerate.

5.5 SVM as the Most Robust Model

SVM was the most consistent classifier across all augmentation types. It was the only model where the Python class maintained 100% precision and recall on every augmentation except cropping. Even on cropping, where overall accuracy dropped to 73.15%, Python still scored 100% on both metrics.

6. Discussion

6.1 Key Observations

Both XGBoost and CNN got near-perfect accuracy on clean images, but their failure modes under augmentation barely overlap. XGBoost handles intensity-based distortions fine but breaks on spatial shifts. The CNN is the other way around. This comes from how each model processes images.

When the CNN failed on bit depth and DPI, it defaulted to predicting Python in both cases. Python/ReportLab documents may have more distinctive binary patterns that survive heavy distortion, which is why the model kept predicting Python even when it was wrong most of the time.

6.2 Limitations

Python only has 100 images vs around 400 for the other two classes. This makes Python-class results less reliable under harder augmentations. All experiments used a single random seed, so how stable these results are across different data splits is not known.

6.3 Future Directions

Training on augmented images would likely improve robustness across the board. Combining SVM and CNN into an ensemble is another option — since they fail on different augmentation types, combining their

predictions could cover each model's weak spots.

7. Conclusion

All four classifiers can tell apart Word, Google Docs, and Python PDF documents from binary-derived grayscale images when the images are clean. Robustness testing showed that does not hold equally well once distortions are introduced.

SVM was the most robust model across augmentation types and the only one to consistently handle the Python class correctly. SGD was a solid second choice. XGBoost and CNN each have one clear failure mode — cropping for XGBoost, bit depth and DPI for CNN.

For real-world use, SVM is the safest general choice given its consistency. Combining SVM and CNN into an ensemble could cover more ground since they fail on different things. Training on augmented data would also help all four models perform better under distortion.